

STATISTICAL MECHANICS



Daniel F. Styer
Oberlin College

Oberlin College
Statistical Mechanics

Daniel F. Styer

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

The LibreTexts name, logo, and book covers are trademarked by LibreTexts, Inc. (a 501(c)(3) not-for-profit organization) and protected against unauthorized use.

This text was compiled on 08/09/2026

- [InfoPage](#) by [Delmar Larsen](#) is licensed [Public Domain](#).

TABLE OF CONTENTS

Licensing

1: The Properties of Matter in Bulk

- 1.1: What is Statistical Mechanics About?
- 1.2: Outline of Book
- 1.3: Fluid Statics
- 1.4: Phase Diagrams
- 1.5: Additional Problems

2: Principles of Statistical Mechanics

- 2.1: Microscopic Description of a Classical System
- 2.2: Macroscopic Description of a Large Equilibrium System
- 2.3: Fundamental Assumption
- 2.4: Statistical Definition of Entropy
- 2.5: Entropy of a Monatomic Ideal Gas
- 2.6: Qualitative Features of Entropy
- 2.7: Using Entropy to Find (Define) Temperature and Pressure
- 2.8: Additional Problems

3: Thermodynamics

- 3.1: Heat and Work
- 3.2: Entropy
- 3.3: Heat Engines
- 3.4: Multivariate Calculus
- 3.5: Thermodynamic Quantities
- 3.6: The Thermodynamic Dance
- 3.7: Non-fluid Systems
- 3.8: Thermodynamics Applied to Fluids
- 3.9: Thermodynamics Applied to Phase Transitions
- 3.10: Thermodynamics Applied to Chemical Reactions
- 3.11: Thermodynamics Applied to Light
- 3.12: Additional Problems

4: Ensembles

- 4.1: The Canonical Ensemble
- 4.2: Meaning of the Term "Ensemble"
- 4.3: Classical Monatomic Ideal Gas
- 4.4: Energy Dispersion in the Canonical Ensemble
- 4.5: Temperature as a Control Variable for Energy (Canonical Ensemble)
- 4.6: The Equivalence of Canonical and Microcanonical Ensembles
- 4.7: The Grand Canonical Ensemble
- 4.8: The Grand Canonical Ensemble in the Thermodynamic Limit
- 4.9: Summary of Major Ensembles
- 4.10: Quantal Statistical Mechanics
- 4.11: Ensemble Problems I
- 4.12: Ensemble Problems II

5: Classical Ideal Gases

- 5.1: Classical Monatomic Ideal Gases
- 5.2: Classical Diatomic Ideal Gases
- 5.3: Heat Capacity of an Ideal Gas
- 5.4: Specific Heat of a Hetero-nuclear Diatomic Ideal Gas
- 5.5: Chemical Reactions Between Gases
- 5.6: Problems

6: Quantal Ideal Gases

- 6.1: Introduction
- 6.2: The Interchange Rule
- 6.3: Quantum Mechanics of Independent Identical Particles
- 6.4: Statistical Mechanics of Independent Identical Particles
- 6.5: Quantum Mechanics of Free Particles
- 6.6: Fermi-Dirac Statistics
- 6.7: Bose-Einstein Statistics
- 6.8: Specific Heat of the Ideal Fermion Gas
- 6.9: Additional Problems
- 6.10: Appendix

7: Harmonic Lattice Vibrations

- 7.1: The Problem
- 7.2: Statistical Mechanics of the Problem
- 7.3: Normal Modes for a One-dimensional Chain
- 7.4: Normal Modes in Three Dimensions
- 7.5: Low-temperature Heat Capacity
- 7.6: More Realistic Models
- 7.7: What is a Phonon?
- 7.8: Additional Problems

8: Interacting Classical Fluids

- 8.1: Introduction
- 8.2: Perturbation Theory
- 8.3: Variational Methods
- 8.4: Distribution Functions
- 8.5: Correlations and Scattering
- 8.6: The Hard Sphere Fluid

9: Strongly Interacting Systems and Phase Transitions

- 9.1: Introduction to Magnetic Systems and Models
- 9.2: Free Energy of the One-Dimensional Ising Model
- 9.3: The Mean-Field Approximation
- 9.4: Correlation Functions in the Ising Model
- 9.5: Computer Simulation
- 9.6: Additional Problems

10: Appendices

- 10.1: A- Series and Integrals
- 10.2: B- Evaluating the Gaussian Integral

- [10.3: C- Clinic on the Gamma Function](#)
- [10.4: D- Volume of a Sphere in d Dimensions](#)
- [10.5: E- Stirling's Approximation](#)
- [10.6: F- The Euler-MacLaurin Formula and Asymptotic Series](#)
- [10.7: G- Ramblings on the Riemann Zeta Function](#)
- [10.8: H- Tutorial on Matrix Diagonalization](#)
- [10.9: I- Catalog of Misconceptions](#)
- [10.10: J- Thermodynamic Master Equations](#)
- [10.11: K- Useful Formulas](#)

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: The Properties of Matter in Bulk

- [1.1: What is Statistical Mechanics About?](#)
- [1.2: Outline of Book](#)
- [1.3: Fluid Statics](#)
- [1.4: Phase Diagrams](#)
- [1.5: Additional Problems](#)

This page titled [1: The Properties of Matter in Bulk](#) is shared under a [CC BY-SA](#) license and was authored, remixed, and/or curated by [Daniel F. Styer](#).

1.1: What is Statistical Mechanics About?

Statistical mechanics treats matter in bulk. While most branches of physics. . . classical mechanics, atomic physics, quantum mechanics, nuclear physics. . . deal with one or two or a few dozen particles, statistical mechanics deals with, typically, about a mole of particles at one time. A mole is 6.02×10^{23} , considerably larger than a few dozen. Let's compare this to a number often considered large, namely the U.S. national debt. This debt is (2014) about 18 trillion dollars, so the national debt is about thirty trillionth of a mole of dollars.¹ Even so, a mole of water molecules occupies only 18 ml or about half a fluid ounce. . . it's just a sip.

The huge number of particles present in the systems studied by statistical mechanics means that the traditional questions of physics are impossible to answer. For example, the traditional question of classical mechanics is the time-development problem: Given the positions and velocities of all the particles now, find out what they will be at some future time. This problem has not been completely solved for three gravitating bodies. . . clearly we will get nowhere asking the same question for 6.02×10^{23} bodies! But in fact, a solution of the time-development problem for a mole of water molecules would be useless even if it could be obtained. Who cares where each molecule is located? No experiment will ever be able to find out. To make progress, we have to ask different questions, question like "How does the pressure change with volume?", "How does the temperature change upon adding particles?", "What is the mean distance between atoms?", or "What is the probability for finding two atoms separated by a given distance?". Thus the challenge of statistical mechanics is two-fold: first find the questions, and only then find the answers.

¹ In contrast, the Milky Way galaxy contains about 0.3 or 0.6 trillionth of a mole of stars. The entire universe probably contains fewer than a mole of stars.

This page titled [1.1: What is Statistical Mechanics About?](#) is shared under a [CC BY-SA](#) license and was authored, remixed, and/or curated by [Daniel F. Styer](#).

1.2: Outline of Book

This book begins with a chapter, *the properties of matter in bulk*, that introduces statistical mechanics and shows why it is so fascinating.

It proceeds to discuss the *principles of statistical mechanics*. The goal of this chapter is to motivate and then produce a conceptual definition for that quantity of central importance: entropy. In contrast to, say, quantum mechanics, it is not useful to cast the foundations of statistical mechanics into a mathematically rigorous “postulate, theorem, proof” mold. Our arguments in this chapter are often heuristic and suggestive; “plausibility arguments” rather than proofs.

Once we have defined entropy and know a few of its properties, what can we do with it? The subject of thermodynamics asks what can be discovered about substance by just knowing that entropy exists, without knowing a formula for it. It is one of the most fascinating fields in all of science, because it produces a large number of dramatic and unexpected results based on this single modest assumption. This book’s chapter on thermodynamics begins by developing a concrete operational definition for entropy, in terms of heat and work, to complement the conceptual definition produced in the previous chapter. It goes on to apply entropy to situations as diverse as fluids, phase transitions, and light.

The chapter on *ensembles* returns to issues of principle, and it produces formulas for the entropy that are considerably easier to apply than the one produced in chapter 2. Armed with these easier formulas, the rest of the book uses them in various applications.

The first three applications are to the classic topics of classical ideal gases, quantal ideal gases, including Fermi-Dirac and Bose-Einstein statistics, and harmonic lattice vibrations or phonons.

The subject of ideal gases (i.e. gases of non-interacting particles) is interesting and often useful, but it clearly does not tell the full story. . . for example, the classical ideal gas can never condense into a liquid, so it cannot show any of the fascinating and practical phenomena of phase transitions. The next chapter treats *weakly interacting fluids*, using the tools of perturbation theory and the variational method. The correlation function is introduced as a valuable tool. This is the first time in the book that we ask questions more detailed than the questions of thermodynamics.

Finally we treat *strongly interacting systems and phase transitions*. Here our emphasis is on magnetic systems. Tools include mean field theory, transfer matrices, correlation functions, and computer simulations. Under this heading fall some of the most interesting questions in all of science. . . some answered, many still open.

The first five chapters (up to and including the chapter on *classical ideal gases*) are essential background to the rest of the book, and they must be treated in the sequence presented. The last four chapters are independent and can be treated in any order.

This page titled [1.2: Outline of Book](#) is shared under a [CC BY-SA](#) license and was authored, remixed, and/or curated by [Daniel F. Styer](#).